SPECIFICATION CHARTS

The specifications provided in this section include screed weights, dimensions, performance, and torque values for both metric and standard inch fastener.

⚠CAUTION Replace original equipment only with components approved by your LeeBoy dealer.

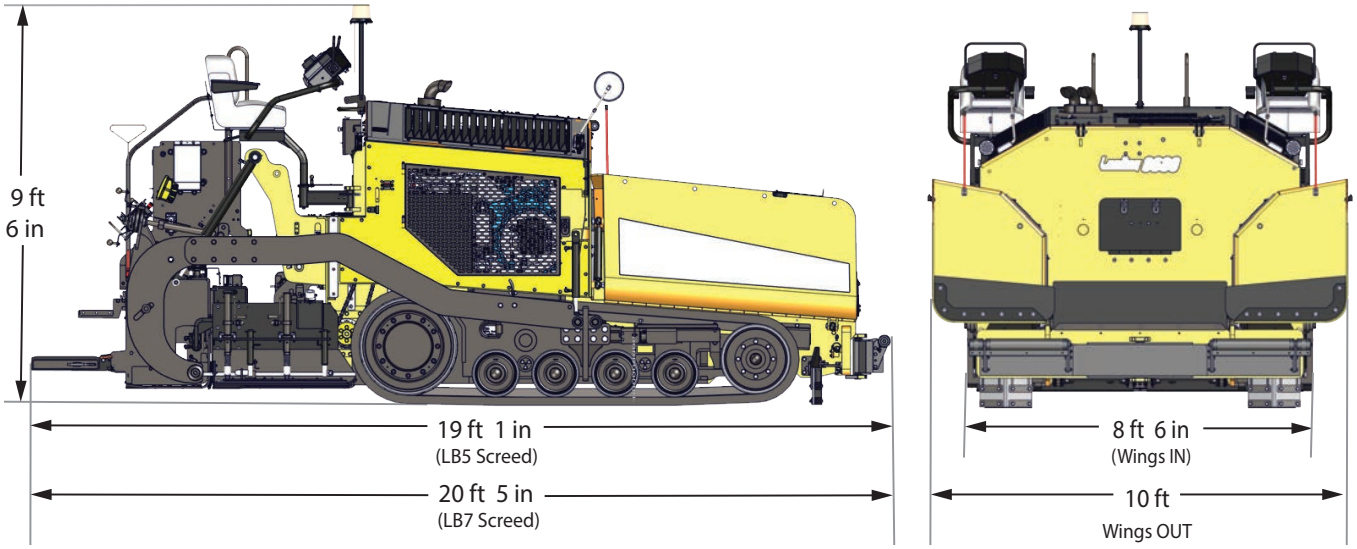


Table 2-1. Machine Dimensions

ITEM	SPECIFICATION
Overall Length	w/LB5 Screed: 19.1 ft w/LB7 Screed: 20.5 ft
Transport Length	w/LB5 Screed: 18 ft 2 in w/LB7 Screed: 19.1 ft
Overall Height	9.6 ft
Transport Height	8.9 ft
Overall Transport Width (with hopper up)	8.6 ft
Overall Width (with hopper down)	10 ft
Total Weight	With LB5 Screed: 33,600 lb With LB7 Screed: 36,000 lb
Ground Clearance	6.25 in

Table 2-2. Kubota Tier 4 Engine Specifications

ITEM	SPECIFICATION
Manufacturer	Kubota
Model	V5009-TIE5
Fuel Type	Diesel
Cylinders	4
High Idle	2200 RPMs
Max Gross Power	210 HP
Max Gross Torque	642 lb ft

Table 2-3. Performance Specifications

ITEM	SPECIFICATION
Travel Speed	0 - 6 mph
Paving Speed	0 - 3.4 mph
Maximum Paving Depth	0 - 10 in (Hydraulic tow point cylinders; includes manual flight screws on screed.)
Turning Radius	15 in (Capable of Counter-Rotation)

Figure 2-4. Lubricant Specifications

ITEM	SPECIFICATION
Engine Oil	15W-40, API, CH-4, CI-4
Hydraulic Oil	All Weather 32
Torque Hub	75W 90 Synthetic Gear Oil
Grease	Shell Avania EP Grease or Equivalent
Chain Lube	Chain Lube

Table 2-5. Machine System Capacity Specifications

ITEM	SPECIFICATION
Hydraulic Oil	37 gal
Fuel	50 gal
Engine Oil with Filter	5 gal
Coolant	7 gal
Spray Down	8 gal
Torque Hub	.88 gal

Table 2-6. Hydraulic System Specifications

ITEM	SPECIFICATION
Drives	Variable volume hydraulic pumps for each drive.
Conveyor/Augers	Load sense pump
Cylinders	Gear pump
Charge/Blower	Tandem gear pump
Primary Filter	Charge Filter (7 Microns)
Secondary Filter	Intank Return Filter (10 micron)
Tank Strainers	Suction Strainers (125 Micron)

Table 2-7. Hydraulic Pressures Specifications

ITEM	SPECIFICATION
Drive	5510 psi
Conveyors/Augers (Load Sense)	3480 psi
Cylinders	3000 psi
Track Tension Pressure	1200 psi
Track Tension Relieving	3600 psi
Standby Pressure	350 psi
Charge Pressure	500 psi

Table 2-8. Electrical Specifications

ITEM	SPECIFICATION
Battery	Two (2), Maintenance-Free
Battery Ampere per Hour Rating	1150 CCA
Battery Voltage	12 V
Alternator Type and Voltage	12 V, Negative Ground
Alternator Output Amperage	150 Amps
Alternator Fan Belt Tension	Automatic belt tension mechanism keeps serpentine belt under tension at all times.
Starter Voltage and Type	12 V, Negative Ground
Starter Rating	4.8 kW

Table 2-9. Generator Specifications

ITEM	SPECIFICATION
Model	Harrison HU34.0ALT-36N-C-5119
Power	34 KW
Voltage	480 V (277 V with 120V option)
Frequency	58 - 62 HZ

Table 2-10. Hopper Specifications

ITEM	SPECIFICATION
Material Capacity	15 Tons
Height	35.5 in
Hopper Wings	Heavy duty, 1/2 inches, with large radius corners, hydraulic lift front apron, and rear hopper wing wipers.

Table 2-11. Augers Specifications

ITEM	SPECIFICATION
System Components	2 independent, hydraulic, segmented reversible augers operate manually or automatically.
Variable Speed	1 - 150 RPMs
Diameter	16 in (or 14-inch option)

Table 2-12. Conveyors Specifications

ITEM	SPECIFICATION
System Components	2 independent, hydraulic, reversible conveyors with automatic shut-off. Heavy-duty chains. Quad bearings on front and rear conveyors.
Width	18 in

Table 2-13. Track Drive Specifications

ITEM	SPECIFICATION
Type	High-speed, smooth rubber with friction drive wheels, oscillating bogey rollers and hydraulically adjustable front idler wheel. Hydrostatically powered.
Track Length	128 in
Track Width	14 in
Number of Speeds	2-speed torque hubs with 2-speed motor consisting of 4 speeds obtained via keypad button.
Ground Contact	99 in each side
Brakes	Fail Safe
Gauge Width (outside tracks)	91.38 in

Table 2-14. LB5 and LB7 Screed Specifications

ITEM	LB5 SCREED	LB7 SCREED (Option)
Screed Weight	5200 lbs	7400 lbs
ScreeD Width	15 ft 6 in	15 ft 6 in
Electric Heat Elements	Two (2) 3000-watt S-Curve elements on main screed. One (1) 1500-watt sealed S-curve element in each screed extension.	Two (2) 3500-watt S-Curve elements on main screed. One (1) 2000-watt sealed blade element in each screed extension.
Main Screed Width	8 ft	8 ft
Screed Extension Width (2 adjustable, hydraulically-operated)	46 in	46 in
Wear Plates	Main Screed Depth: 18 in Extension Depth: 9 in	Main Screed Depth: 24 in Extension Depth: 10 in
Wear Plate Thickness	Main Screed Depth: .375 in Extension Depth: .375 in	Main Screed Depth: .50 in Extension Depth: .50 in
Bullnose Radius	Main Screed: 1.688 in Screed Extension: 1.688 in	Main Screed: 2.5 in Screed Extension: 2.5 in
Vibration	Two (2) vibration motors producing 4100 vpm	Two (2) vibration motors producing 4100 vpm
Slope @ Maximum Width	3.5 in	4 in
Crown/Valley @ 10 ft	Positive Crown: 2.5 in Negative Crown: 1.5 in	Positive Crown: 2.5 in Negative Crown: 1.5 in
End Gate	Heated, spring-loaded with vertical height adjustments and tilt.	Heated, spring-loaded with vertical height adjustments and tilt.

Figure 2-15. Push Rollers Specifications

ITEM	SPECIFICATION
Components	Two (2) 3.5 inch (diameter) adjustable rollers with sealed bearings on swivel frame and self-cleaning scrapers.
Vertical Height Adjustment	7 in

Figure 2-16. Spray Down Specifications

ITEM	SPECIFICATION
Components	Electric heavy-duty pump, with 2 hoses, sight gauge and spray wand.
Citrus Tank Capacity	8 gal

TORQUE SPECIFICATIONS

The following tables list torque values for standard hardware. This is a guide for average application involving typical stresses and machined surfaces. Values are based upon physical limitations of clean, plated and lubricated hardware. Under more extreme conditions, individual torque value should be followed.

Conversion formulas are provided below:

Conversion	Formula
ft-lb to N•m	[ft-lb]*1.3558 = [N•m)
ft-lb to in-lb	[ft-lb]*12 = [in-lb]
N•m to in-lb	[N•m]*8.8508 = [in-lb]

Standard Inch Fasteners

Table 2-17. Torque Specifications For Standard Inch Fasteners

SIZE	THREAD	CAPSCREWS: SAE GRADE 5				CAPSCREWS: SAE GRADE 8			
		TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
		Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
1/4	20 UNC	8	6	11	8	12	9	16	12
	28 UNF	10	7	14	9	14	10	19	14
5/16	18 UNC	17	13	23	18	25	18	34	24
	24 UNF	19	15	26	20	27	20	37	27
3/8	16 UN C	31	23	42	31	44	33	60	45
	24 UNF	35	26	47	35	49	37	66	50
7/16	14 UNC	49	37	66	50	70	52	95	71
	20 UNF	55	41	75	56	78	58	106	79
1/2	13 UNC	75	57	102	77	106	80	144	108
	20 UNF	85	64	115	87	120	90	163	122
9/16	12 UNC	109	82	1 48	111	154	115	209	156
	18 UNF	121	91	164	123	171	128	232	174
5/8	11 UNC	150	113	203	153	212	159	287	216
	18 UNF	170	127	230	172	240	180	325	244
3/4	10 UNC	267	200	362	271	376	282	510	382
	16 UNF	297	223	403	302	420	315	569	427
7/8	9 UNC	429	322	582	437	606	455	822	617
	14 UNF	474	355	643	481	669	502	907	681
1	8 UNC	644	483	873	655	909	681	1232	923
	14 UNF	722	542	979	735	1020	765	1383	1037
1-1/4	7 UNC	1121	840	1520	1139	1817	1363	2464	1848
	12 UNF	1241	930	1683	1261	2012	1509	2728	2046
1-1/2	6 UNC	1950	1462	2644	1982	3162	2371	4287	3215
	12 UNF	2194	1645	2975	2230	3557	2668	4823	3617

Metric Fasteners

Table 2-18. Torque Specifications for Metric Fasteners

NOMINAL SIZE AND PITCH	CLASS 8.8 [GRADE 5 EQUIVALENT]				CLASS 10.9 [GRADE 8 EQUIVALENT]			
	TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
	Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
M4 x 0.7	2	2	3	2	3	2	4	3
M5 x 0.8	5	3	7	4	7	5	9	7
M6 x 1	8	6	11	8	11	8	15	11
M8 x 1.25	19	14	26	19	27	20	37	27
M10 x 1.5	37	28	50	38	53	40	72	54
M12 x 1.75	65	49	88	66	93	70	126	95
M14 x 2	104	78	141	106	148	111	201	150
M16 x 2	161	121	218	164	230	173	312	235
M18 x 2.5	222	167	301	226	318	239	431	324
M20 x 2.5	314	236	426	320	449	337	609	457
M22 x 2.5	428	321	580	435	613	460	831	624
M24 x 3	543	407	736	552	777	582	1053	789
M27 x 3	796	597	1079	809	1139	854	1544	1158
M30 x 3.5	1079	809	1463	1097	1544	1158	2093	1570

Hydraulic Fittings

Tightening Flare-Type Tube Fittings

1. Check the flare and flare seat for defects that might cause leakage.
2. Align tube with fitting before tightening.
3. Lubricate connection.
4. Hand tighten swivel nut until snug.
5. To prevent twisting the tube(s), use two wrenches. Place one wrench on the connector body and tighten the swivel nut with the second to the torque shown in the following table:

NOTE: The torque values shown are based upon lubricated connections.

Table 2-19. Torque Specifications for Steel Flare Type Tube Fittings

TUBE SIZE OUTER DIAMETER	NUT SIZE ACROSS FLATS	TORQUE VALUE	
(IN)	(IN)	(LB FT)	(N•m)
3/16	7/16	8	11
1/4	9/16	12	16
5/16	5/8	16	22
3/8	11/16	23	31
1/2	7/8	38	52
5/8	1	54	73
3/4	1 1/4	75	102
7/8	1 3/8	83	113

Determining Proper Torque

The only reliable method of creating a consistently leak-free and long-lasting connection is to ensure the coupling is brought to the proper torque. Using a torque wrench with crowfoot is the best method, but the flats method can be used if a torque wrench is not available.

The most straightforward method of determining the correct torque setting is to multiply the desired torque by the length of the wrench from the center of the handle to the center of the drive (L); divided by the length of the wrench from the center of the handle to the crowfoot center (LA) as shown below:

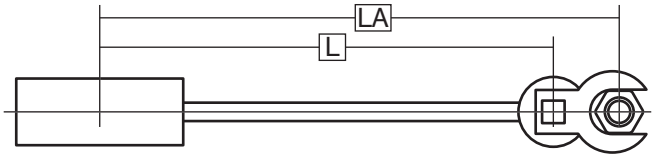


Figure 2-2. Torque Wrench - Crowfoot

NOTICE **The minimum torque values are adequate for sealing most applications. Maximum torque values should never be exceeded.**

There are several methods of determining the correct setting on the torque wrench when using a crowfoot. All of the methods involve making the setting proportional to the effective change in length of the wrench multiplied by the desired final torque. The equations and illustration below describe proper measurements.

Equations

- Torque setting if the crowfoot is placed in line with respect to the wrench:

TS = TD * L / LA

or

TS = TD * L / (L+E)

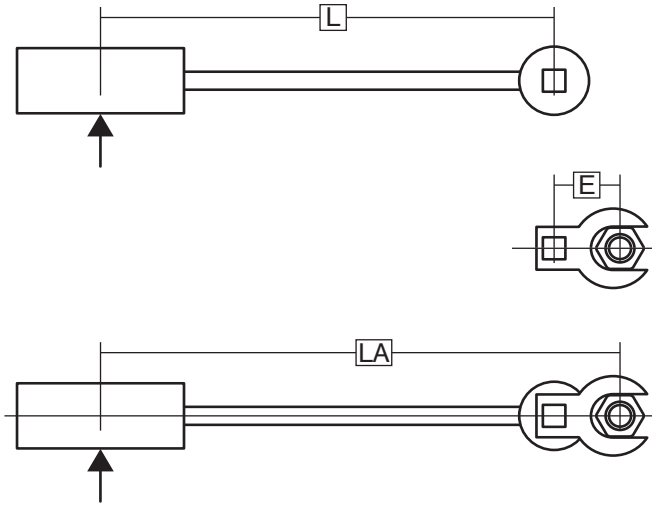


Figure 2-3. Measurements Needed

LEGEND

L = Distance from center of torque wrench handle to the center of socket drive

E = Distance from center of socket drive to the center of crowfoot

LA = Distance from center of torque wrench handle to the center of crowfoot

TD = Desired torque at the fitting

TS = Torque setting indicated on wrench

2

Coupling Installation

Use the following steps for proper coupling installation:

1. Determine the correct torque value for your coupling.

NOTE: Only use the torque values specified from the manufacturer. DO NOT use SAE torque recommendations.

2. Ensure the seal face and threads are clean and in good condition. O-Rings should be lubricated with light oil, but threads should be completely dry unless making pipe thread connections (interference seal).

NOTE: Attach the male end of the hose onto the equipment first since it may be necessary to rotate the entire hose assembly to tighten the male threads. Then route the hose into position while avoiding twisting the hose.

3. Tighten the connection (by hand), bringing the seal face into contact and rotating the nut until it stops.
4. Mark a line across the coupling nut and backup hex for the flats method verification of coupling torque.
5. Apply a wrench to the backup hex to prevent the coupling and hose from moving while tightening the nut with a torque wrench.

NOTICE **Failure to retain the backup hex during installation will also result in additional clamp load force that could cause damage to the seal face.**

NOTE: The coupling nut must be in motion for an accurate torque reading. If the nut is stopped before final torque value is achieved, it must be loosened and retightened until the torque is attained while the nut is in motion.

If a torque wrench cannot fit into the coupling area or if it is unavailable, the flats method may be used to ensure that the coupling is properly tightened, as shown in the following figure.

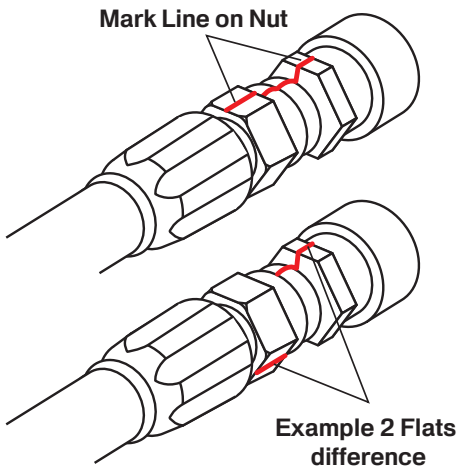


Figure 2-20.

Figure 2-21. Flats Method Tightening

NOTE: The mark placed on the nut and backup hex after tightening by hand should rotate during final tightening according to the table below. The nut and backup hex can then be marked to indicate if the coupling loosens over time.

Table 2-22. Flats Method Values for Selected Terminations

FLATS METHOD VALUES		
Termination Type	**Dash Size**	**Flats**
JIC	-4	1.5 - 1.75
JIC	-6	1.0 - 1.5
JIC	-8	1.5 - 1.75
JIC	-10	1.0 - 1.5
JIC	-12	1.0 - 1.5
JIC	-16	.75 - 1.0
JIC	-20	.75 - 1.0
JIC	-24	.75 - 1.0
JIC	-32	.75 - 1.0
JIS	-4	.5 - 1.5

1. Seal faces must be in contact with the fitting fully tightened by hand before marking flats.
2. The flats method is most accurate for the first assembly cycle. For multiple disassembly and assembly cycles, torque values are more reliable.
3. Tightening two (2) flats or more may damage seal faces.

Table 2-23. Torque Specifications For US Style Coupling Terminations

JIC, SAE 45°, ORFS, O-RING BOSS, GATES ADAPTERLESS AND MEGASEAL										
DASH SIZE	JIC 37°, SAE 45° & Mega-Seal (Steel)		JIC 37°, SAE 45° & Mega-Seal (Brass)		Flat Face O-Ring Seal (Steel)		SAE O-Ring Boss (Steel) & Gates Adapterless ≤ 4000 PSI		SAE O-Ring Boss (Steel) & Gates Adapterless > 4000 PSI	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
-3									8	10
-4	10	11	5	6	10	12	14	16	14	16
-5	13	15	7	9					18	20
-6	17	19	12	15	18	20	24	26	24	26
-8	34	38	20	24	32	40	37	44	50	60
-10	50	56	34	40	46	56	50	60	72	80
-12	70	78	53	60	65	80	75	83	125	135
-14					65	80			160	180
-16	94	104	74	82	92	105	111	125	200	220
-20	124	138	75	83	125	140	133	152	210	280
-24	156	173	79	87	150	180	156	184	270	360
-32	219	243	158	175						

Table 2-24. Torque Specifications for DIN 24, DIN 60, and Inverted Cone Style Coupling Terminations

DIN 24, DIN 60, AND INVERTED CONE			
Size (mm)		Torque (lb ft)	
Light Series Tube OD	Heavy Series Tube OD	Min	Max
6		7	15
8		15	26
10	8	18	30
12	10	22	33
14	12	26	37
15	14	30	52
	16	30	52
18	20	44	74
22	25	59	89
28	30	74	111
	38	74	162
35		133	184
42		148	221

Table 2-25. Torque Specifications for 4-Bolt Flange Connections

Figure 2-26. 4-BOLT FLANGES		
Dash Size	Bolt Size (in)	Torque (lb ft)
-8	0.31	17
-12	0.38	26
-16	0.44	43
-20	0.50	65
-24	0.63	130
-32	0.75	220

1. Align faces and tighten bolts (by hand) before applying final torque in a pattern. The seal faces must be parallel with an even bolt tension to seal properly.
2. Torque values apply to bolts that are plated or coated in light engine oil.
3. Before assembly, lubricate O-Ring with light oil (SAE 10W or 20W).

Table 2-27. Torque Specifications for NPTF Dry Seal Pipe Threads

Figure 2-28. NPTF	
Dash Size	**Max Torque (ft-lb)**
-2	20
-4	25
-6	35
-8	45
-12	55
-16	65
-20	80
-24	95
-32	120

1. The torque values obtained from tightening pipe threads can vary considerably depending upon thread condition. Adequate sealing can occur at values much lower than the maximum values listed above. Only enough torque to achieve adequate sealing should be used.
2. When using a male tapered pipe thread with a female straight or parallel pipe thread, maximum values are 50% of those listed in the table above.
3. If thread sealant is used, maximum values shown should be decreased by 25%.

Table 2-29. Torque Specifications for BSP 30° Inverted Cone and JIS Coupling Terminations

BSP 30° INVERTED CONE AND JIS		
Dash Size	**Torque (ft-lb)**	
	Min	**Max**
-2	7	9
-4	11	18
-6	19	28
-8	30	36
-10	37	44
-12	50	60
-16	79	95
-20	127	152
-24	167	190
-32	262	314

Section 3
COMPONENT LOCATION

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NOTES



COMPONENTS OVERVIEW

The LeeBoy 8680 Commercial Paver delivers higher horsepower and production capacity for a larger footprint in the heavy commercial class market. It's designed and built for producing superior mat quality in a variety of road and other pavement construction projects.

Dual main operator control platforms with ergonomic control keypads include auto-throttle, auto-idle and pause functions. Adjustable operator platforms that swing out beyond the frame to provide optimum visibility with the 8680 commercial paver. The beacon light can also be lowered as needed for overhead clearance.

The rugged 8680 commercial paver features high-speed, smooth rubber tracks with oscillating bogies on a four-speed drive system. This paver can travel up to 6 mph with a paving speed of 300 feet per minute (3.4 mph) due to the rubber tracks and a 210 HP engine. Continuous and flexible, the rubber track provides the traction and flotation of a tracked paver, but with better mobility and a smoother ride.

The 8680 features the LB5 screed or the optional LB7 screed with the ability for variable paving widths from 8 to 15.5 feet. Independent control of one or both hopper wings in its 9-ton hopper allows paving alongside buildings or other obstructions with ease. Narrow conveyors and a power tunnel shields the height-adjustable augers (14 or 16-inch) are just part of LeeBoy innovation and added versatility. Proportional control of augers and conveyors enhance asphalt material delivery.

Separate pumps for each function charge, (augers, blower and conveyors), cylinders and propulsion, with proportional valves for extensions and tow cylinders to optimize hydraulic performance. Hydraulic tow point cylinders allows 10 inches of travel for quick and easy height adjustments, and integrates with sonic grade and slope automation. Sloping extensions include 2.5-inch chrome rods with multiple adjustments for improved stability.

Operator Platforms



Dual operator control platforms slide out from either side of the paver for better visibility and feature the latest in technology. Each fully-contained control platform features modern amenities such as a comfortable seat, foot rest, and an ergonomic digital control panel. The stylish control platform mobility combined with its low-profile hood line gives operators unlimited visibility with no line-of-sight disruptions while paving.

The controls are always directly in front of the operator as the entire platform can be adjusted to multiple height positions. The air cleaner and emissions system is mounted below the low-profile hood line to reduce overall height and enhance the operator's ability to see and communicate with the material truck driver. The beacon light can easily be lowered as needed for overhead clearance.

Its sleek, robust design and innovative control panel include push-button keypad controls for paving functions. Operators can choose paving functions at a glance of symbols on the control panel keypad buttons, which also illuminate to show function status. The Plus+ 1® CAN bus system is the brains behind paver controls. The main control panel features the DM430 digital display controller, work function switches, and a vandalism cover for security and protection. The display's wide-angle, anti-glare screen automatically adjusts brightness for sunlight or darker settings. The operator can set the desired paving speed in the display controller and the automated system will pave consistently throughout the day.



High-Speed Rubber Tracks

Rubber tracks offer a smooth, less noisy ride at faster speeds with maximum flotation and more stability, uniformity and consistency. This rubber tracked paver is extremely helpful when paving close to buildings or other tight areas as the turning radius is tighter--you can turn it around with itself on counter-rotation.

Rubber tracks also have an advantage when paving wide and/or deep lifts as there is no variation in machine movements when pushing trucks. The 8680 paver rubber tracks combines a stronger tractive effort than steel tracks, with the smoother ride of a wheeled paver.

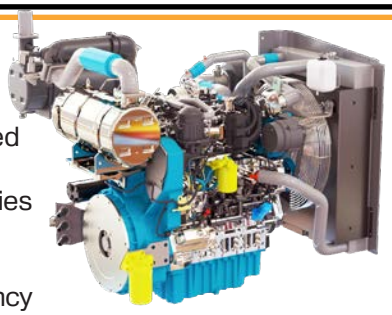
Hydraulic System

Separate pumps for the various functions enhance the paver's hydraulic power. The hydraulic system consists of a tandem drive pump, a tandem blower and charge pump, cylinder pump; and a load-sense pump for the augers and conveyors. Hydraulic motors and cylinders that work together for optimum machine component control. Valves regulate the directional, pressure and fluid flow throughout the hydraulic system. Manifolds are set up to allow for all options. Screed function manifolds are mounted onto the screed. Dual 220,000 BTU oil coolers improve reliability. Hydraulic pressure transducers sense hydraulic pressure levels and send output signals to the control panel digital display unit for ease and convenience when checking and troubleshooting main functions.

Electrical System

The two batteries produces 12 volts DC and maintains 950 cold cranking amperes (CCA) each. An engine-mounted alternator with 150 amperes charging capacity keeps the batteries charged during normal operation.

Engine



A powerful, turbo-charged 210 HP engine equipped with the latest technologies and advanced emission systems allows optimum performance and efficiency while operating functions simultaneously. Cooling systems increase water pump flow while robust lubrication systems produce more flow with higher pressure.

This powerful engine features a remotely-mounted DOC+DPF+SCR after-treatment package with more flow, high pressure, enhanced cooling system and heat reduction. A heat shield for the exhaust manifold reduces heat radiation to the engine. Baffles and louvers direct hot air up and away from the operator. Six rib serpentine belt and pulleys with P-Max fan adds more air flow through the engine cooling system.

Advanced emission-control devices and low-sulfur fuel requirements--only Ultra-Low Sulfur Diesel (ULSD) can be used in these engines--combine with after-treatment Diesel Particulate Filter (DPF) to optimize combustion. This paver is also equipped with a DEF (Diesel Exhaust Fluid) tank. Engine optimization runs all functions at 1800 RPMs for fuel-efficient operation.

The airflow design draws ambient air across the engine compartment, allowing the exhaust air to exit from the side of the machine to provide a cooler engine compartment and operator platform. This also reduces engine power demand, maximizes fuel efficiency, minimizes emissions and enhances quiet operation.

Accessing the radiator and other engine components is as simple as pushing a switch to raise and lower the engine hood. The engine is also accessible through several access doors, with a front service door where fuel filter components are located for easy access.

Refer to the Engine Operator's Manual accompanying your paver for additional engine information.

Screed

The 8680 Commercial Paver features the new LB5 screed that is packed with standard features along with other custom options. Or choose the monumental LB7 screed option, packed with easy-to-use highway paving features especially for screed operators. Both provide variable paving widths from 8 to 15.5 feet, horizontal depth screws, automatic temperature control, and spring-loaded end gates for an unblemished asphalt mat.



Multiple adjustment mechanisms are located on top of the screed, extensions and end gate along with adjustable strike-offs. Electric dual vibrators mounted on the main screed frame increase paving material compaction. The screed extensions also include slope and berm capability. The LB7 screed option also includes electronic vertical height and angle-of-attack (AOA) adjustments, along with hydraulic crown/invert adjustment. An energy chain on each screed extension protects the hard wiring as the screed extension moves back and forth for added protection and less maintenance. S-curve heating elements in the main screed with straight heating elements in the extensions cycle during heat-up to save horsepower and fuel consumption while paving.



Augers

Right and left augers rotate to move material from the conveyors to the screed. The 14 or 16-inch augers can be adjusted up or down, and operated independently of each other.

The augers are controlled from the operator platform control panel or by the screed operator using the screed control box on each side of the paver. Operators can control the augers manually or automatically using the sonic sensor mounted on either screed end gate, where it monitors and controls the pile height of material at the discharge end of the augers.

The dual augers are hardened, 3/4-inch thick, 14 or 16-inch round, HRC 55-60 augers, that can operate in reverse. The modular auger assembly, in one-foot sections, allows for a simple quick change of the assembly.

Hopper and Conveyors

Material in the hopper is moved toward the back of the paver to the screed by conveyors.

The narrow, automatic conveyors with a wide hopper design delivers asphalt more evenly and efficiently to produce a smooth asphalt mat. The conveyors are also automatic, adjustable, and reversible through the screed control boxes.

Hydraulic power raises and lowers the hopper wings, which can also be operated individually while paving close to buildings, barriers and other obstacles. The hopper's rounded design reduces cold material build-up while rear hopper wing wipers help prevent spillage. The hydraulic front apron can be raised and lowered to form a snug fit with the material truck, which also helps clean asphalt off the apron and front corners of the hopper. The hopper wings can also be locked into place (in the closed position) for transport.



MACHINE OVERVIEW

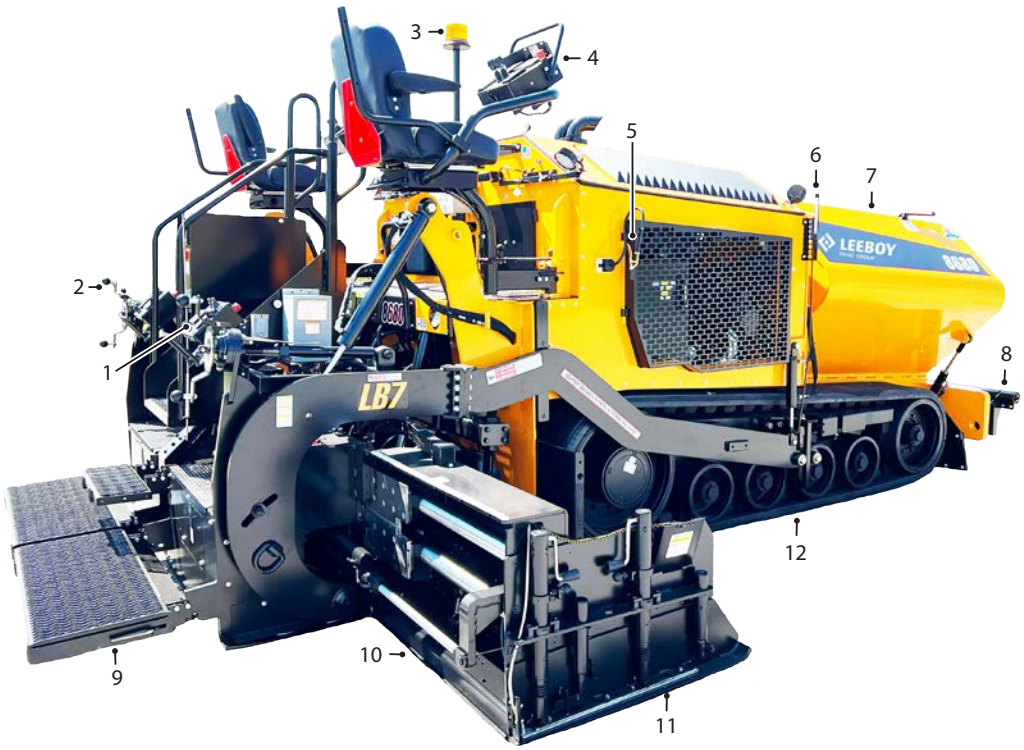


Figure 3-1. Machine Overview

ITEM NO.	CONTROL NAME	FUNCTION
1	Screed Control Panel	Located on each side of the screed. Features numerous paving controls.
2	Horizontal Depth Screw	Controls depth of the asphalt (one on each side).
3	Beacon Light	Revolving amber safety light to alert traffic and others at the work site.
4	Main Control Panel	Main operator control panel with joysticks, optional steering wheel or optional steering knob.
5	Spray Down Wand	Spray wands located on both sides for daily cleaning after paving.
6	Screed Elevation Guide	This gauge provides the operator with a visual reference for the tow point's position (one on each side).
7	Hopper	Hopper wings fold out to hold asphalt material.
8	Push Rollers or Hydraulic Truck Hitch Option	Push rollers adjusts height (manually) for roll-up to the asphalt material truck. The optional truck hitch is hydraulically driven for roll-up, but also hooks around the inside of the asphalt truck wheel, hitching the paver to the truck. Controlled from the main control panel.
9	Foldable Walkboard	Walkboard provides added convenience for screed operators. Can be folded up when not in use.
10	Screed Extension	Extends paving width from 8 to 15.5 feet (one extension on each side).
11	End Gate	Adjusts for a paving width to match joints and leave a clean edge.
12	High-Speed Rubber Tracks	Offers a smoother and more quiet ride with better traction than steel tracks.

MAIN CONTROL PANEL OVERVIEW

Left Control Panel Shown



Figure 3-2. Main Control Panel Overview

Table 3-1. Main Control Panel Overview

ITEM NO.	CONTROL NAME	FUNCTION
1	Lower Left Keypad	The lower left keypad features 8 buttons and contains the following paver functions: Left and Right Auger/Conveyor On/Off; Left and Right Cut-Off Open/Closed; and Left Screed Extension Extend/Retract.
2	Upper Left Keypad	The upper left keypad features 12 buttons and contains the following paver functions: Start/Stop (On/Off); Beacon/Work Lights; Grade/Slope; Truck Hitch In/Out; Spray Down On/Off; Slope/Berm Select; Slope/Berm Up/Down; Front Apron Up/Down and Vibrator On/Off.
3	DM430 Digital Display	Displays engine operating information, status and system faults.
4	Upper Right Keypad	The upper right keypad features 12 buttons and contains the following paver functions: Work High/Low Speed; Travel High-Low Speed; Horn; Left and Right Tow Up/Down; Auger Lift Up/Down; Pause; Apron Up/Down; Pad Builder; and Generator On/Off.
5	E-Stop Button	Immediately stops the paver engine and all electronic functions. Turn clockwise to release E-Stop button.
6	Lower Right Keypad	The lower right keypad features 8 buttons and contains the following paver functions: Hopper Wing Select; Active Console; Hopper Wing(s) Open/Close; Screed Raise; Screed Float; and Right Screed Extension Extend/Retract.
7	Joysticks	Joysticks control speed and direction forward or reverse, and also turns the paver right or left. The farther pushed in either direction, the faster the speed. The locking collar locks the joystick in NEUTRAL. You must pull up on the locking collar to move joysticks forward or reverse.
8	Steering Knob Option	If equipped with this option, the steering knob assembly replaces the joystick on the right side.

CONTROL PANEL WITH STEERING WHEEL OPTION

The control icons and keypads are the same, but arranged slightly different if equipped with the steering wheel option.

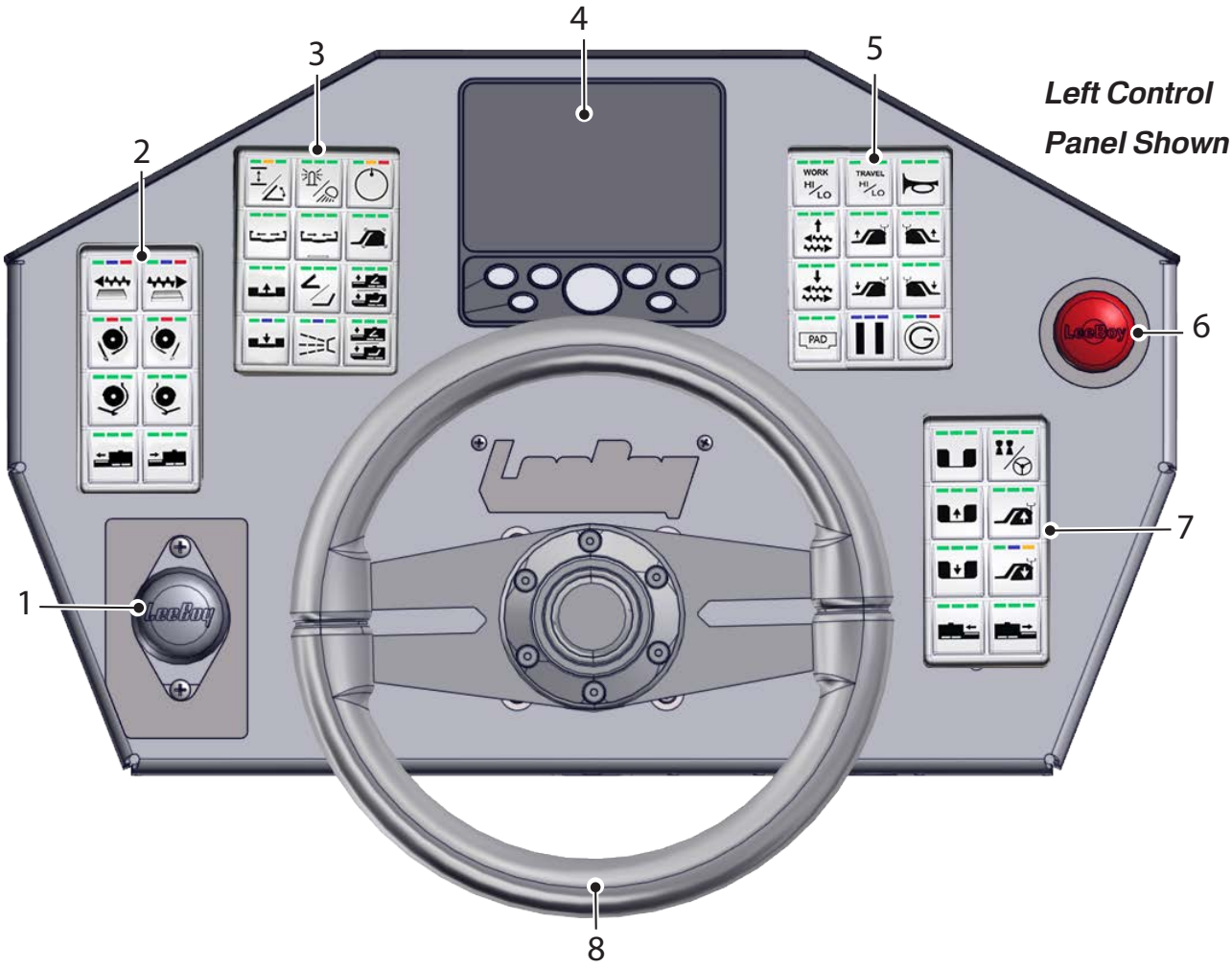


Figure 3-3. Main Control Panel With Steering Wheel Overview

Table 3-1. Main Control Panel Overview

ITEM NO.	CONTROL NAME	FUNCTION
1	Forward/Neutral/ Reverse (FNR) Joystick	Joysticks control speed and direction by pushing the joystick forward or reverse. The center position is neutral. The farther pushed in either direction, the faster the speed. The locking collar locks the joystick in NEUTRAL. You must pull up on the locking collar to move joysticks forward or reverse.
2	Lower Left Keypad	The lower left keypad features 8 buttons and contains the following paver functions: Left and Right Auger/Conveyor On/Off; Left and Right Cut-Off Open/Closed; and Left Screed Extension Extend/Retract.
3	Upper Left Keypad	The upper left keypad features 12 buttons and contains the following paver functions: Start/Stop (On/Off); Beacon/Work Lights; Grade/Slope; Truck Hitch In/Out; Spray Down On/Off; Slope/Berm Select; Slope/Berm Up/Down; Front Apron Up/Down and Vibrator On/Off.
4	DM430 Digital Display	Displays engine operating information, status and system faults.
5	Upper Right Keypad	The upper right keypad features 12 buttons and contains the following paver functions: Work High/Low Speed; Travel High-Low Speed; Horn; Left and Right Tow Up/Down; Auger Lift Up/Down; Pause; Apron Up/Down; Pad Builder; and Generator On/Off.
6	E-Stop Button	Immediately stops the paver engine and all electronic functions. Turn clockwise to release E-Stop button.
7	Lower Right Keypad	The lower right keypad features 8 buttons and contains the following paver functions: Hopper Wing Select; Active Console; Hopper Wing(s) Open/Close; Screed Raise; Screed Float; and Right Screed Extension Extend/Retract.
8	Steering Wheel	Steers right and left.

Upper Left Keypad

(Left Operator Platform Control Panel Shown)

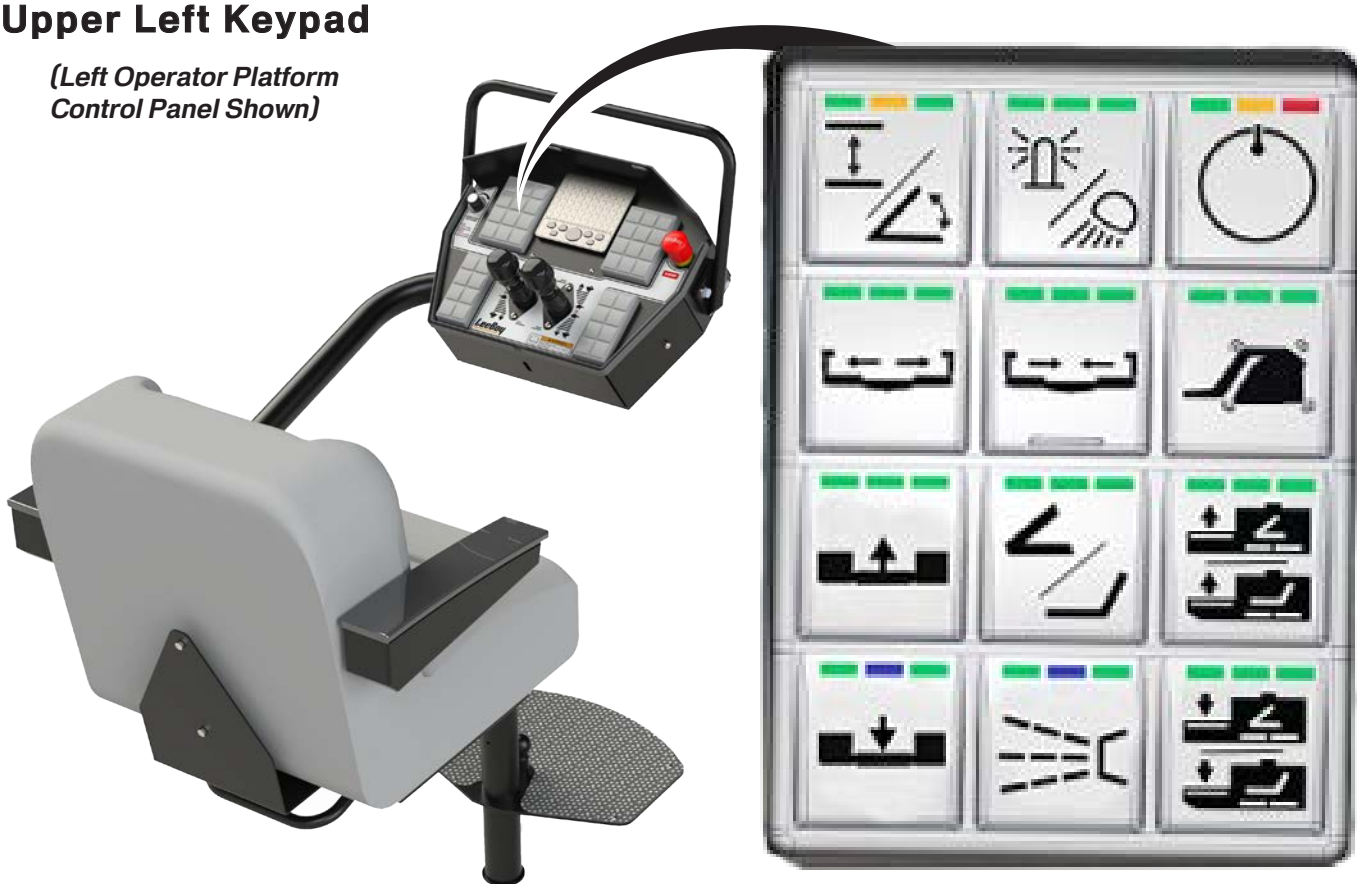
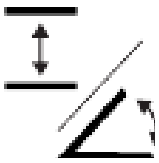
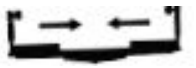
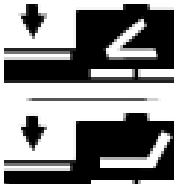
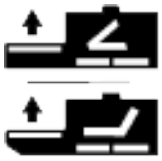


Table 3-2. Control Panel - Upper Left Keypad

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Grade/Slope	You can choose to control tow points manually or automatically via the TopCon® sonic automatic grade/slope control. The default setting is MANUAL. The left LED light illuminates GREEN in this mode on both consoles. Press and hold the button to set up the TopCon® sensor system and operate in AUTO mode. The right LED light illuminates GREEN in this mode on both consoles. **NOTE: The Grade/Slope function is disabled when joysticks are in neutral. The center light will illuminate YELLOW.**
	Beacon/Work Lights	Upon start-up, the lights are OFF (default) and can be operated from the active and inactive consoles. LED status lights also illuminate on both consoles. Press the button once to activate the beacon light (the left LED light illuminates GREEN). Press the button again to activate the work lights (the center LED light illuminates GREEN). Press the button a third time to activate the beacon and work lights (all three LED lights illuminate GREEN). Pressing the button a fourth time turns all lights OFF.
	Start/Stop (On/Off)	This button allows the operator to start and stop the engine from the active or inactive console. With ignition key in the ON position and all four joysticks in neutral, press this button and hold one second to start the engine. The center light illuminates amber during “Wait to Start” stage. The left LED lights illuminate GREEN on both consoles when the engine is running. Push the button again to stop the engine. The right LED light illuminates RED. **NOTE: All joysticks on BOTH operator platform control panels *must* be in NEUTRAL and the E-Stop button engaged to start the engine.**

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Truck Hitch OUT (OPTION)	Attaches truck hitch to the material truck if equipped with this option. Can be operated from active and inactive consoles (LED status lights Illuminate GREEN while engaged on both consoles). Press and hold button to move truck hitch arms outward.
	Truck Hitch IN (OPTION)	Attaches truck hitch to the material truck if equipped with this option. Can be operated from active and inactive consoles (LED status lights Illuminate GREEN while engaged on both consoles). Press and hold button to move truck hitch arms inward.
	Vibrator	This button turns the screed vibrator ON and OFF and can be operated from active or inactive consoles. The screed vibrator helps compact material while paving. Press the button once to turn ON. All three LED lights will illuminate green on both consoles when engaged. The vibrator automatically turns OFF if paver is placed in reverse or neutral.
	Apron UP	Use this button to raise the hopper's front apron to help clean asphalt off the apron and front corners of the hopper. Press and hold button to desired angle. The LED status lights Illuminate GREEN while engaged on both consoles.
	Apron Down	Use this button to lower the hopper's front apron of the hopper to help clean asphalt off the apron and front corners of the hopper. Press and hold button to desired angle. The LED status lights Illuminate GREEN while engaged on both consoles.
	Slope/Berm Select	Use this button to select SLOPE or BERM mode. Press the button once to select SLOPE. The left LED light illuminates GREEN in this mode on both consoles and the left or right screed keypad (corresponding to the console activated). Push the button twice to select BERM mode. The right LED light illuminates GREEN on both consoles and the corresponding screed keypad.
	LEFT Slope/Berm DOWN	Pressing this button enables the Slope or Berm DOWN function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both consoles and the left or right screed keypad (corresponding to the console activated).
	LEFT Slope/Berm UP	Pressing this button enables the Slope or Berm UP function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both consoles and the left or right screed keypad (corresponding to the console activated).
	Spray Down	Engages the spray down system for spraying release agent to clean, lubricate and prevent asphalt from hardening on machine surfaces. Press the button once to turn ON. All three LED lights will illuminate GREEN. Press the button again to turn OFF. The spray down function can be engaged on both active and inactive consoles.

Lower Left Keypad

(Left Operator Platform Control Panel Shown)

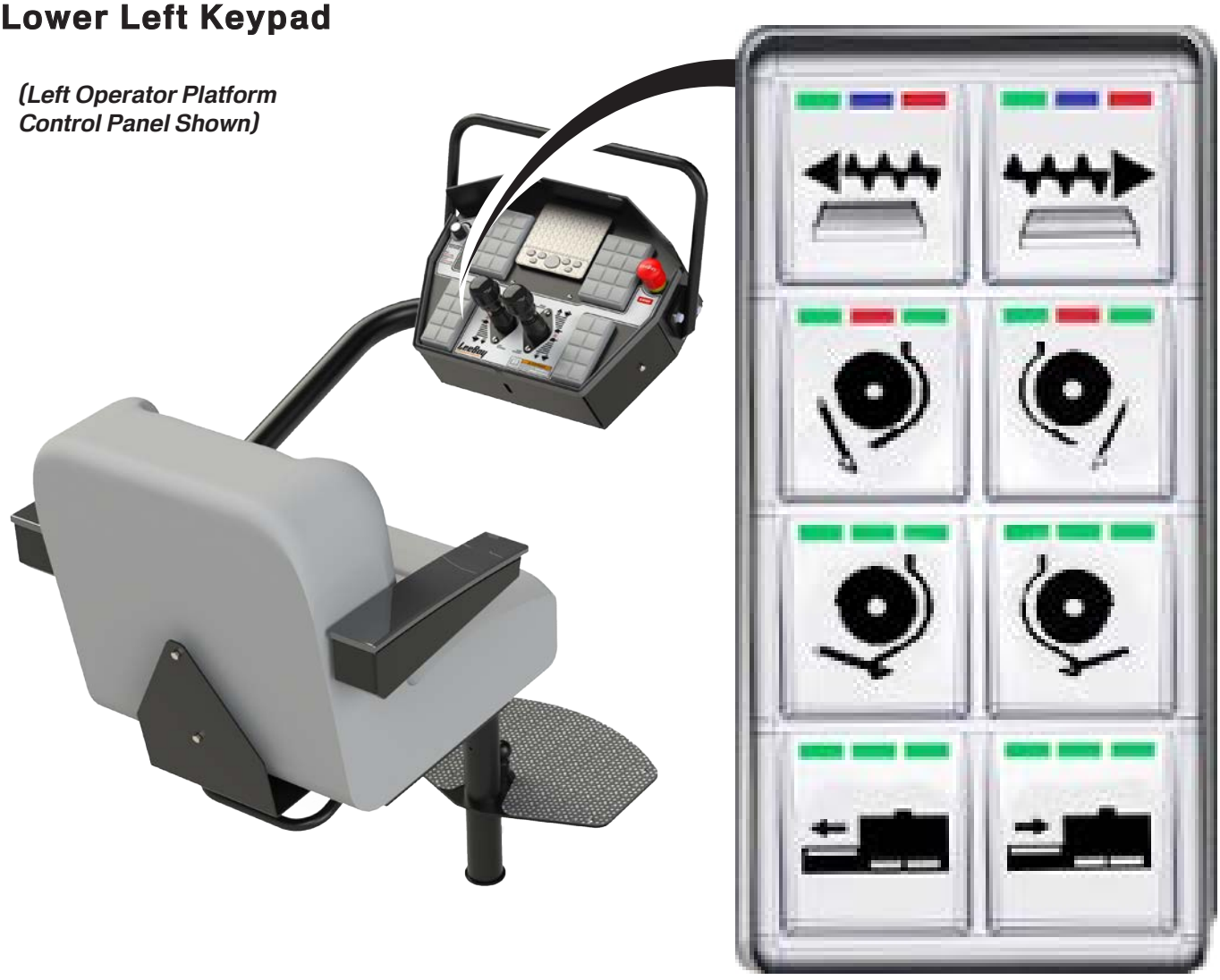


Table 3-3. Control Panel - Lower Left Keypad

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Auger/Conveyor LEFT	Turns the LEFT auger and conveyor on and off. The left LED light illuminates GREEN while conveyor is ON; the right RED light illuminates when OFF, and BLUE if paver is paused while conveyors are operating (or the material has shut off the sensor). Press the button once to turn ON the LEFT auger and conveyor (AUTO mode). Press the button and hold for MANUAL override. Press button again to turn off the LEFT auger and conveyor. **NOTE: Moving the joysticks into neutral pauses operation in AUTO mode.**